A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Forest resources & deforestation - Set 1

Mark scheme

AQA spec 3.5.3; Biological resources; textbook pp. 360-369

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1 Timber:** building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) **1 Fuel:** fuelwood and charcoal - still the main energy source for most people in LEDCs **1 Food:** fruit, nuts, bushmeat, livestock fodder – crops of forest origin (coffee, cacao, bananas, Brazil nuts) – pigs and chickens were forest animals **1 Fibres:** paper (wood pulp cellulose), cotton, viscose / rayon **1 Medicines:** quinine (cinchona), taxol (yew), aspirin (willow) – most species not yet researched **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time – total forest carbon (vegetation plus soil) is about double that in the atmosphere **1** Growing trees sequester CO_2 by photosynthesis and release oxygen – forest soils hold more carbon than the trees (especially boreal) **1** Planting / protecting forests removes CO_2 and reduces the enhanced greenhouse effect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **enhanced greenhouse effect** → extra warming caused by added greenhouse gases **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Highest transpiration of any ecosystem increases downwind rainfall – interception increases evaporation – deep organic soils retain water and delay runoff so river flow is more even and flooding reduced **1** Low albedo absorbs sunlight – heat stored and released at night reduces day-night temperature extremes – canopy microclimate: lower light, wind, higher humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	

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Question	Marking guidance	Marks
02.1	**Marks shared** Up to 6 marks in total - share them between the 5 points below according to how completely each one is made. **up to 2** Non-indigenous species (Sitka spruce from North America) chosen for fast growth and pest resistance – indigenous wildlife's needs for food, pollination, seed dispersal and habitat are not met **up to 2 Single species:** easy planting, thinning and felling over large areas – but each tree species supports its own community so few species overall **up to 2** Close planting makes trees compete for light so they grow tall and straight (good timber) but shades out ground flora and the species that depend on them **up to 2 Simple age structure:** uniform size makes management easy but gives little variety of light, temperature, wind – no old trees or dead wood for invertebrates and woodpeckers **up to 2 Mixed woodland:** varied niches, stratification, dead wood, indigenous trees each with their own dependent species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Trees are felled when income will be greatest - approaching full size when the growth rate starts to decline **1** Continuing to grow them ties up land and capital for little extra timber – natural forests keep old and dead trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Industrial society replaced species-specific uses (baskets, fencing, charcoal, tool handles) with large-scale uses: chipboard, plywood, pulp **1** Wood fuel replaced by fossil fuels / electricity – so traditional management declined and woods became uniform **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 365-367; AQA spec 3.5.3.2.	

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Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 3 points below. **1 South America:** high loss (5.1-5.5 Mha yr^{-1}) falling by about half by 2020 **1 Africa:** loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest **1 Europe:** negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
03.2	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 South America:** clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export – HEP reservoirs – roads improving access – mining **up to 2 Africa:** growing rural populations clearing land for subsistence farming – fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy) – fodder cutting in dry areas **up to 2 Both:** hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted – urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **HEP** → electricity generated from falling water **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** They create barriers fragmenting wildlife populations **1** Improved access makes it easier to remove timber and farm produce, so clearance spreads along them **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	1 mark for the point below. **1** Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades – slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Less dead organic matter / leaf litter so less humus and slower soil formation – fewer nutrients returned **1** Loss of canopy and litter exposes soil to rain splash and wind **1** Decomposing roots no longer bind soil so erosion, gullying and landslides increase (especially on slopes) **1** Reduced organic matter and biota lower infiltration and water retention **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **humus** → dark, decomposed organic matter in soil • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	4
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind **1 Increased runoff:** more flooding after rain, lower flows in dry periods – sedimentation of rivers **1** Increased albedo of cleared land (reflects more sunlight) **1** Reduced carbon sequestration and loss of the carbon reservoir (CO_2 released by burning / decay) contributing to climate change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
04.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects **1** Loss of inter-species relationships (pollinators, seed dispersers) causing further declines **1** Loss of genetic resources / potential medicines before they are discovered **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
05.1	**1 mark each** 1 mark for each of the 5 points below. **1** Harvest rate below the maximum sustainable yield – only economically valuable species (mahogany) selectively logged, not clear-felled **1** Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing – large seed trees and small trees left to grow **1** Buffer zones with no harvesting or machinery near rivers / wetlands **1** Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees **1** Each felled tree tagged and tracked to the retailer for FSC accreditation – rangers control illegal logging – tree nurseries replant over-exploited and ecologically important species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it **1** Employment as rangers, forestry and tourism workers – ecotourists use local restaurants, produce and souvenirs – offcuts given to local carvers **1** Benefits give the community an incentive to protect the forest – carbon sequestration payments from companies fund management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept – timber resource used over a longer period **1 Disadvantage:** more labour-intensive and expensive – more tracks needed per tree **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	
06.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
06.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** Local people had used the area for fuelwood, timber for house building, clearings for farming, hunting and the collection of medicines and fibres, all of which protection could have stopped **1** Local people are employed as rangers, forestry workers and in ecotourism, so the forest is worth more to them standing than cleared **1** Ecotourist groups eat in local restaurants, are fed on local produce and buy locally made souvenirs, spreading the income through the community **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Trees are only felled where other trees of the same species stand upwind of the clearing, so their seed will blow in and recolonise it **1** Small trees are left standing so that they can grow and be harvested in the future when they are larger **1** Selected large trees are left because they produce many seeds that fill the gaps left by the trees that were felled **1** No mechanical or harvesting activity takes place near rivers, because aquatic ecosystems are ecologically sensitive **1** The use of large machines is restricted because they damage the trunks and roots of the trees that are not being felled **1** Track routes are planned to avoid important trees and their roots, including rare trees and those that feed monkeys or birds, and old tracks are re-used **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	4
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** The tag allows the timber to be tracked from the stump where it grew to its final retail destination, such as a DIY store in the UK **1** That chain of custody is the evidence the FSC needs that the wood really came from a sustainably managed forest and was not mixed with illegally logged timber **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
07.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1 Ecotourism:** visitors pay to see the forest and its wildlife, and local businesses supply them **1** Timber confiscated from illegal loggers by the rangers is sold, and the money pays for conservation work **1** Electricity companies, airlines and individuals fund an area of forest as a carbon sequestration research project, because they want to offset atmospheric carbon dioxide **1** Lower-value species such as zericote are developed for furniture, musical instruments and carving, and small branches and offcuts are given to local artisans **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
07.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The canopy is only opened in small gaps, so the forest microclimate - shade, humidity and temperature - is maintained and shade-dependent species survive **1** The soil keeps its cover of vegetation and leaf litter, and the roots that bind it, so erosion, leaching and sedimentation of rivers are far lower **1** Seed trees and small trees remain, so the gaps are recolonised by the same species and the forest regenerates naturally without replanting **1** Habitats and food sources are left in place and the forest is not fragmented, so populations of birds and mammals are not isolated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leaching** → nutrients or ions washed down out of the soil / washed away by water	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369, 386-387; AQA spec 3.5.3.3.	

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